

Spacecraft Proximity Operations Under Motion and Input Constraints: A Learning-Based Robust Optimal Control Approach

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This article proposes a learning-based robust control scheme for spacecraft proximity operations under motion constraints (i.e., approaching path and sensor field-of-view constraints), input saturation, and external disturbances. To enhance the robustness of the learning algorithm, a disturbance observer with finite-time convergence is first designed to provide accurate model information for online learning. By virtue of the barrier functions and hyperbolic tangent functions, a performance index is developed, which incorporates both motion and input constraints into the framework of adaptive dynamic programming. Then, following the actor-critic structure, an approximate optimal saturated control policy is obtained using two neural networks (NNs), wherein the weights of the NNs are updated online. It is shown that the derived controller can guarantee the boundedness of system states and network weight estimation errors, while ensuring the satisfaction of motion and input constraints despite the presence of external disturbances. Finally, numerical simulations are carried out for spacecraft proximity operations with a tumbling target to verify the effectiveness of our proposed method.

I. INTRODUCTION

On-orbit servicing (OOS) has received wide attention over the past decades owing to its extensive applications, such as assembling large space structures, removing orbital debris, and refueling a powerless satellite [1], [2], [3]. Several space programs have been conducted for on-orbit demonstration of OOS technologies, including

NASA's Restore-L [4], DARPA's Orbital Express [5], ESA's ClearSpace-1 [6], JAXA's ETS-VII [7], etc. Spacecraft proximity operations, as a key technology of OOS, aim to steer the servicing spacecraft to safely arrive at the docking/berthing position with a certain attitude. During proximity operations, the servicing spacecraft is required to simultaneously perform the relative position and attitude motions with respect to the passive target (e.g., defunct satellite, space debris, etc.). This induces the six-degree-of-freedom (6-DOF) pose control problem, for which various control methods have been presented in the literature [8], [9], [10], [11], just to name a few. However, most of these existing methods cannot balance control cost and performance and may result in large fuel consumption. To overcome this problem, some scholars proposed optimal control methods, which can extend the on-orbit lifespan of servicing spacecraft by performance and energy optimization.

Model predictive control and reinforcement learning (RL) are two major methods for optimal control of dynamical systems [12]. The former usually needs to numerically solve the optimal control problems, thus requiring large computational resources. While the latter, also known as adaptive dynamic programming (ADP) in the control community, experienced a process from offline iteration to online update realization [13]. The online version of ADP algorithms can derive a near-optimal control policy by using neural networks (NNs) to approximate the cost function online, which is conducive to real-time implementation. More specifically, the weights of NNs are updated online using the real-time observation data along the system trajectory. In this manner, the approximation error can be reduced and the initial controller is improved by using an online adaption/learning algorithm. Different from

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offline training that produces an optimal controller after iterative learning, this type of algorithm can be regarded as an online training method since the controller is updated continuously online and learns optimality as the system runs over time. Survey papers on the ADP-based optimal control can be found in [14], [15], and [16]. Vamvoudakis and Lewis [17] proposed a policy iteration scheme for learning the continuous-time optimal control solution of nonlinear systems, which guarantees convergence of the learning algorithm and stability of the closed-loop system. In recent years, the ADP-based control architecture has been successfully applied to various systems, such as spacecraft [18], robotic systems [19], unmanned aerial vehicles [20], etc. Despite recent advances in ADP-based optimal control, its applications to spacecraft proximity operations still face two critical problems, namely, multiconstraint and robustness.

The pose control design for spacecraft proximity operations should account for various constraints, which can be generally categorized into two types: 1) motion constraints and 2) input saturation. In order to accomplish the proximity operations safely, the servicing spacecraft is required to comply with some motion constraints, among which the most important two types are approaching path constraint and sensor field-of-view (FOV) constraint. Specifically, the servicing spacecraft should approach the desired docking/berthing position within a certain safe corridor to avoid collision with some target components, which is referred to as the approaching path constraint, while the sensor FOV constraint arises from the requirement of real-time relative navigation, that is, the detectable zone of visual sensors must always cover the target to provide real-time relative state information for the servicing spacecraft. The existing methods for the aforementioned motion constraints include path planning methods and artificial potential function (APF) methods. In addition, input saturation caused by the physical limitation of actuators (e.g., reaction wheels, control momentum gyros, and thrusters) also poses great challenges for the pose control design, and several results have been reported in the literature [21], [22], [23]. The ADP-based methods provide promising solutions to the pose control problem of spacecraft proximity operations under multiple constraints. In this respect, Dong et al. [24] proposed an ADP-based controller for spacecraft attitude reorientation with explicit consideration of attitude and angular rate constraints. Hu et al. [25] designed a learning-based control method for the pose control of spacecraft while dealing with both path and FOV constraints. It is noted, however, the authors in [24] and [25] do not account for input saturation. Recently, Yang et al. [26] developed an ADP-based saturated attitude control scheme with prescribed performance for spacecraft subject to input saturation. Wei et al. [27] proposed a novel RL-based gain tuning method, which can reduce the control amplitude effectively. Nonetheless, it is still an open problem to design an ADP-based control scheme for spacecraft proximity operations under both motion and input constraints.

External disturbances are another issue to be considered, which may degrade control precision. Since the ADP

technique is usually model based, its robustness is of great importance for antidisturbance concerns. Passive robust control accounts for the worst condition and relies on the robustness of its own algorithm. For example, in [28], a robust optimal controller is proposed for spacecraft attitude stabilization under actuator misalignments and disturbances. Its robustness is achieved by adding an antidisturbance term to the traditional optimal control. Zhao et al. [29] investigated the ADP-based robust control problem against actuator faults, using the learning control policy to construct a fault-tolerant controller. On the other hand, active robust control uses measurement or estimation information to compensate for disturbances. Modares et al. [30] presented an adaptive optimal control method, which can deal with system uncertainties using an NN-based identifier. Meng et al. [31] designed a disturbance observer (DOB) to estimate the synthetic uncertainty/disturbance term, and then, used the estimation information to solve the modified optimal control problem. However, the optimality of these existing ADP-based robust controllers is somehow reduced due to the transformation from a traditional control policy to a robust one. Moreover, input saturation remains a tricky problem for ADP-based robust control. Thus, it is a non-trivial task to simultaneously enhance the antidisturbance and constraint-handling abilities of the ADP algorithms.

Motivated by the aforementioned challenges, this article presents an ADP-based robust optimal control scheme for spacecraft proximity operations under motion and input constraints, as well as external disturbances. First, the relative position and attitude dynamics with a thruster-only actuation system are established for spacecraft proximity operations with a tumbling target. The 6-DOF error dynamics is then written into a simpler generalized form such that the optimal control theory can be easily applied. A disturbance observer with finite-time convergence is developed to obtain an accurate estimation of external disturbances, which is used for online learning to enhance robustness. An especially designed performance index is proposed to cope with motion and input constraints. In particular, a nonquadratic control-related term is used to derive a saturated control law, and two barrier functions are constructed to encode path and FOV constraints. Subsequently, an ADP-based approximation algorithm is proposed to derive an online learning control policy for optimal spacecraft pose control. The main contributions of this article are twofold.

- 1) A nonquadratic term and two barrier functions are tactfully introduced into the performance index, whereby the derived control policy can deal with both input and motion constraints.
- 2) The DOB technique is incorporated into ADP-based control law design to improve its robustness, and thus, the network weights are updated based on a more accurate system model.

The rest of this article is organized as follows. Section II provides the dynamic model for spacecraft proximity operations and describes environmental disturbances, motion constraints, and thruster configuration. The main results are

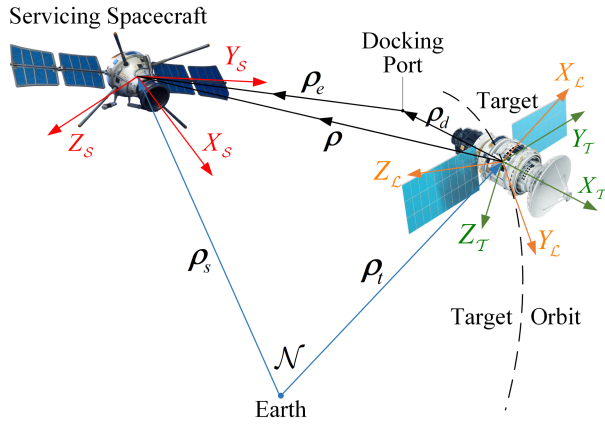


Fig. 1. Coordinate frame systems.

presented in Section III, where a saturated ADP-based control scheme is designed based on a finite-time disturbance observer. Simulation results are shown in Section IV, and finally, Section V concludes this article.

Notations: Throughout this article, $\mathbb{R}^n$ is the n -dimensional Euclidean space, $\mathbf{I}_n$ is the $n \times n$ identity matrix, and $\mathbf{0}_{m \times n}$ is the $m \times n$ zero matrix. $\lambda_m(\cdot)$ denotes the minimum eigenvalue of a square matrix, $\|\cdot\|$ denotes the two-norm of a vector or a matrix, and $\mathbf{R}_{AB}$ denotes the rotation matrix from the frame $\mathcal{B}$ to $\mathcal{A}$. The unit quaternion set is denoted as $\mathbb{Q}_u = \{\mathbf{q} = [\mathbf{q}_v^\top, q_4]^\top \in \mathbb{R}^4 \mid \mathbf{q}^\top \mathbf{q} = 1\}$, and the notation “ $\odot$ ” is the quaternion multiplication operator. In addition, $\mathbf{S}(\mathbf{x}) = [0, -x_3, x_2; x_3, 0, -x_1; -x_2, x_1, 0]$ gives a skew-symmetric matrix for any vector $\mathbf{x} = [x_1, x_2, x_3]^\top$.

II. PRELIMINARIES

Four coordinate frames are defined to describe the relative motion of the servicing spacecraft with respect to a freely tumbling target, as shown in Fig. 1, where $\mathcal{N}$ is the Earth-centered inertial frame, $\mathcal{L}$ is the local-vertical-local-horizontal frame, while $\mathcal{S}$ and $\mathcal{T}$ denote the body frames of the servicing spacecraft and target, respectively.

A. Relative Position Dynamics

Define $\mathbf{r}^{\mathcal{N}} = \mathbf{r}_s^{\mathcal{N}} - \mathbf{r}_t^{\mathcal{N}}$ as the relative position between the servicing spacecraft and target expressed in $\mathcal{N}$, with $\mathbf{r}_s^{\mathcal{N}} \in \mathbb{R}^3$ and $\mathbf{r}_t^{\mathcal{N}} \in \mathbb{R}^3$ being the radius vectors from the center of Earth to the servicing spacecraft and target, respectively. Then, the relative position motion of the servicing spacecraft with respect to the target in $\mathcal{N}$ is described as

$$\ddot{\mathbf{r}}^{\mathcal{N}} = -\mu \left(\frac{\mathbf{r}_s^{\mathcal{N}}}{r_s^3} - \frac{\mathbf{r}_t^{\mathcal{N}}}{r_t^3} \right) + \frac{\mathbf{f}^{\mathcal{N}}}{m_s} \quad (1)$$

where $\mu = 398600.4418 \text{ kg}^3/\text{m}^2$ is the geocentric gravitational constant, r_s and r_t are the length of radius vectors $\mathbf{r}_s^{\mathcal{N}}$ and $\mathbf{r}_t^{\mathcal{N}}$, $\mathbf{f}^{\mathcal{N}} \in \mathbb{R}^3$ is the control force expressed in $\mathcal{N}$, and m_s is the mass of the servicing spacecraft. In practical engineering, the relative position dynamics is usually established in $\mathcal{S}$ for the convenience of relative state measurement. Denote by $\boldsymbol{\rho}$, $\boldsymbol{\rho}_s$, and $\boldsymbol{\rho}_t$ the mappings of $\mathbf{r}^{\mathcal{N}}$, $\mathbf{r}_s^{\mathcal{N}}$, and $\mathbf{r}_t^{\mathcal{N}}$ in $\mathcal{S}$, respectively. Then, it follows that $\boldsymbol{\rho} = \mathbf{R}_{\mathcal{S}\mathcal{N}} \mathbf{r}^{\mathcal{N}}$.

Differentiating $\boldsymbol{\rho}$ and using $\dot{\mathbf{R}}_{\mathcal{S}\mathcal{N}} = -\mathbf{S}(\boldsymbol{\omega}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}}) \mathbf{R}_{\mathcal{S}\mathcal{N}}$ ($\boldsymbol{\omega}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}}$ is the inertial angular velocity of the servicing spacecraft expressed in $\mathcal{S}$), we have

$$\dot{\mathbf{v}} = \dot{\boldsymbol{\rho}} + \boldsymbol{\omega}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}} \times \boldsymbol{\rho} \quad (2)$$

where $\mathbf{v} = \mathbf{R}_{\mathcal{S}\mathcal{N}} \dot{\mathbf{r}}^{\mathcal{N}}$ is the relative velocity of the servicing spacecraft with respect to the target expressed in $\mathcal{S}$, and the second term on the right-hand side of (2) is the so-called connected velocity. Based on (2), the second-order derivative of $\boldsymbol{\rho}$ is further evaluated as

$$\ddot{\boldsymbol{\rho}} = -\dot{\boldsymbol{\omega}}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}} \times \boldsymbol{\rho} - 2\boldsymbol{\omega}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}} \times \dot{\boldsymbol{\rho}} - \boldsymbol{\omega}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}} \times \boldsymbol{\omega}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}} \times \boldsymbol{\rho} + \mathbf{R}_{\mathcal{S}\mathcal{N}} \ddot{\mathbf{r}}^{\mathcal{N}} \quad (3)$$

where the first three terms on the right-hand side of (3) can be viewed as the tangential, Coriolis, and radial acceleration, respectively. Substituting (1) into (3) and further considering a disturbance force $\mathbf{f}_d \in \mathbb{R}^3$, it yields that

$$\mathbf{M} \ddot{\boldsymbol{\rho}} + \mathbf{C} \dot{\boldsymbol{\rho}} + \mathbf{D} \boldsymbol{\rho} + \mathbf{w}_1 = \mathbf{f} + \mathbf{f}_d \quad (4)$$

in which $\mathbf{M} = m_s \mathbf{I}_3$, $\mathbf{C} = 2m_s \mathbf{S}(\boldsymbol{\omega}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}})$, $\mathbf{D} = m_s [\mathbf{S}(\dot{\boldsymbol{\omega}}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}}) + \mathbf{S}^2(\boldsymbol{\omega}_{\mathcal{S}\mathcal{N}}^{\mathcal{S}}) + (\mu/r_s^3) \mathbf{I}_3]$, $\mathbf{w}_1 = m_s \mu (\boldsymbol{\rho}_t/r_s^3 - \boldsymbol{\rho}_t/r_t^3)$, and $\mathbf{f} \in \mathbb{R}^3$ is the control force exerting on the servicing spacecraft in $\mathcal{S}$. The radius of target orbit is evolved as $\rho_t = a_o(1 - e_o^2)/(1 + e_o \cos v_o)$, where a_o , e_o , and v_o are the semimajor axis, orbital eccentricity, and true anomaly of the target orbit, respectively. Then, one has $\boldsymbol{\rho}_t = \mathbf{R}_{\mathcal{L}\mathcal{S}}^\top [\rho_t, 0, 0]^\top$, where $\mathbf{R}_{\mathcal{L}\mathcal{S}} = \mathbf{R}_{\mathcal{L}\mathcal{N}} \mathbf{R}_{\mathcal{N}\mathcal{S}}^\top$ with $\mathbf{R}_{\mathcal{L}\mathcal{N}}$ given by [11]

$$\mathbf{R}_{\mathcal{L}\mathcal{N}} = \begin{bmatrix} c_{\Omega_o} c_{\theta_o} - s_{\Omega_o} s_{\theta_o} c_{i_o} & s_{\Omega_o} c_{\theta_o} + c_{\Omega_o} s_{\theta_o} c_{i_o} & s_{\theta_o} s_{i_o} \\ -c_{\Omega_o} s_{\theta_o} - s_{\Omega_o} c_{\theta_o} c_{i_o} & -s_{\Omega_o} c_{\theta_o} c_{i_o} & c_{\theta_o} s_{i_o} \\ s_{\Omega_o} s_{i_o} & -c_{\Omega_o} s_{i_o} & c_{i_o} \end{bmatrix}$$

where $s_\star \triangleq \sin(\star)$ and $c_\star \triangleq \cos(\star)$; $\theta_o = \omega_o + v_o$ is the argument of latitude; and Ω_o , ω_o , and i_o are the right ascension of the ascending node, argument of perigee, and orbit inclination, respectively.

Similar to [32] and [33], the disturbance force is modeled as $\mathbf{f}_d = \mathbf{f}_d^{J_2} + \mathbf{f}_d^o$, where $\mathbf{f}_d^{J_2}$ denotes the J_2 perturbing force, and $\mathbf{f}_d^o$ denotes other external forces. In particular, $\mathbf{f}_d^{J_2} = \mathbf{R}_{\mathcal{S}\mathcal{N}} \boldsymbol{\alpha}_{J_2}^{\mathcal{N}}$ with

$$\boldsymbol{\alpha}_{J_2}^{\mathcal{N}} = -\frac{3\mu J_2 R_e^2}{2\|\mathbf{r}_s\|^5} (\mathbf{U} - 5(z_s/\|\mathbf{r}_s\|)^2 \mathbf{I}_3) \mathbf{r}_s \quad (5)$$

where $J_2 = 0.0010826267$, $R_e = 6378.137 \text{ km}$, $\mathbf{r}_s = \mathbf{R}_{\mathcal{S}\mathcal{N}}^\top \boldsymbol{\rho}_s$, $\mathbf{U} = \text{diag}[1, 1, 3]$, and z_s is the third element of $\mathbf{r}_s$.

During the proximity operations, the servicing spacecraft is required to move towards a docking/berthing port, which we assume, without loss of generality, is located along the $-x$ -axis of $\mathcal{T}$. Denote $\mathbf{p}_d = [p_d, 0, 0]^\top$ ($p_d < 0$) as the anchoring point expressed in $\mathcal{T}$, and thus, $\boldsymbol{\rho}_d = \mathbf{R}_{\mathcal{S}\mathcal{T}} \mathbf{p}_d$ is the desired position in $\mathcal{S}$. As shown in Fig. 1, the relative position tracking error is written as $\boldsymbol{\rho}_e = \boldsymbol{\rho} - \boldsymbol{\rho}_d$. Based on (4), the position tracking error dynamics is given by

$$\mathbf{M} \ddot{\mathbf{v}}_e + \mathbf{C} \dot{\mathbf{v}}_e + \mathbf{G} = \mathbf{f} + \mathbf{f}_d \quad (6)$$

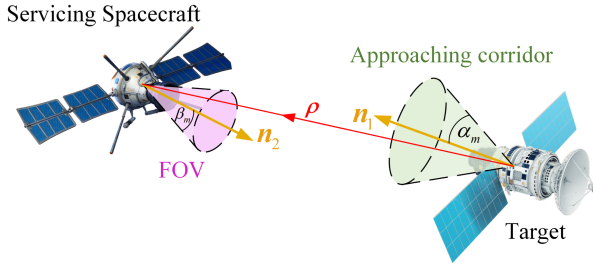


Fig. 2. Illustration of motion constraints.

where $\mathbf{v}_e = \dot{\boldsymbol{\rho}}_e$, $\mathbf{G} = \mathbf{D}\boldsymbol{\rho} + \mathbf{w}_1 + \mathbf{w}_2$, and $\mathbf{w}_2 = \mathbf{M}(\boldsymbol{\omega}_e \times \boldsymbol{\omega}_e \times \boldsymbol{\rho}_d - \dot{\boldsymbol{\omega}}_e \times \boldsymbol{\rho}_d) - \mathbf{C}(\boldsymbol{\omega}_e \times \boldsymbol{\rho}_d)$ with $\boldsymbol{\omega}_e = \boldsymbol{\omega}_{S/N}^S - \mathbf{R}_{S/T}\boldsymbol{\omega}_{T/N}^T$ being the angular velocity error and $\boldsymbol{\omega}_{T/N}^T$ being the inertial angular velocity of target expressed in $\mathcal{T}$.

To avoid collision with some target components, the servicing spacecraft should approach the anchoring point within a safe corridor, as shown in Fig. 2. To ensure this constraint, the relative position should satisfy

$$\alpha = \arccos\left(\frac{\boldsymbol{\rho}}{\|\boldsymbol{\rho}\|} \cdot \mathbf{R}_{S/T}\mathbf{n}_1\right) < \alpha_m \quad (7)$$

where α is the angle between the unit vector of relative position and $\mathbf{n}_1$ in $\mathcal{S}$, $\mathbf{n}_1 = [-1, 0, 0]^T$ is the unit axis vector of conic approaching corridor expressed in $\mathcal{T}$, and α_m is the conic half-angle.

B. Relative Attitude Dynamics

A free-tumbling target is investigated in this article, and its attitude dynamics is given by [34]

$$\dot{\mathbf{q}}_t = \frac{1}{2}\mathbf{Q}(\mathbf{q}_t)\boldsymbol{\omega}_{T/N}^T, \quad \mathbf{Q}(\mathbf{q}_t) = \begin{bmatrix} S(\mathbf{q}_{tv}) + q_{t4}\mathbf{I}_3 \\ -\mathbf{q}_{tv}^T \end{bmatrix} \quad (8)$$

$$\mathbf{J}_t\dot{\boldsymbol{\omega}}_{T/N}^T = -\mathbf{S}(\boldsymbol{\omega}_{T/N}^T)\mathbf{J}_t\boldsymbol{\omega}_{T/N}^T \quad (9)$$

where $\mathbf{q}_t \in \mathbb{Q}_u$ describes the attitude of target, and $\mathbf{J}_t \in \mathbb{R}^{3 \times 3}$ is the inertia of target. Furthermore, based on the attitude motions of servicing spacecraft, the dynamics of attitude tracking error can be described by

$$\dot{\mathbf{q}}_e = \frac{1}{2}\mathbf{Q}(\mathbf{q}_e)\boldsymbol{\omega}_e \quad (10)$$

$$\mathbf{J}_s\dot{\boldsymbol{\omega}}_e = -\mathbf{S}(\boldsymbol{\omega}_{S/N}^S)\mathbf{J}_s\boldsymbol{\omega}_{S/N}^S + \mathbf{J}_s[\mathbf{S}(\boldsymbol{\omega}_e)\mathbf{R}_{S/T}\boldsymbol{\omega}_{T/N}^T - \mathbf{R}_{S/T}\dot{\boldsymbol{\omega}}_{T/N}^T] + \boldsymbol{\tau} + \boldsymbol{\tau}_d \quad (11)$$

where $\mathbf{q}_e = \mathbf{q}_t^{-1} \odot \mathbf{q}_s$ is attitude tracking error with $\mathbf{q}_s \in \mathbb{Q}_u$ representing the attitude of the servicing spacecraft. In addition, $\mathbf{J}_s \in \mathbb{R}^{3 \times 3}$ is the inertia matrix of the servicing spacecraft, and $\boldsymbol{\tau} \in \mathbb{R}^3$ and $\boldsymbol{\tau}_d \in \mathbb{R}^3$ are control and disturbance torques exerting on the servicing spacecraft in $\mathcal{S}$.

Similar to [32] and [33], the disturbance torques are modeled as $\boldsymbol{\tau}_d = \boldsymbol{\tau}_d^g + \boldsymbol{\tau}_d^o$ with the gravity-gradient torque given by

$$\boldsymbol{\tau}_d^g = 3\mu \frac{\mathbf{S}(\boldsymbol{\rho}_s)\mathbf{J}_t\boldsymbol{\rho}_s}{\|\boldsymbol{\rho}_s\|^5} \quad (12)$$

and $\boldsymbol{\tau}_d^o$ is considered to cover all neglected external torques.

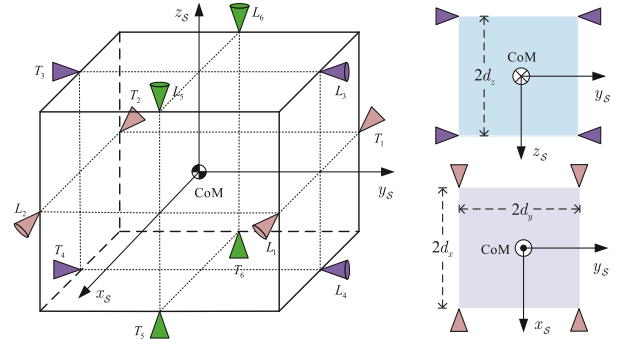


Fig. 3. Thruster configuration.

To obtain real-time relative navigation information, it is essential for the servicing spacecraft to keep the target in sight, as shown in Fig. 2. This corresponds to the FOV constraint, which can be formulated as

$$\beta = \arccos\left(-\frac{\boldsymbol{\rho}}{\|\boldsymbol{\rho}\|} \cdot \mathbf{n}_2\right) < \beta_m \quad (13)$$

where β is the angle between the reverse unit vector of relative position and $\mathbf{n}_2$, $\mathbf{n}_2 = [1, 0, 0]^T$ is the unit boresight vector of the conic FOV described in $\mathcal{S}$, and β_m is the conic half-angle.

C. Thruster Configuration

In this article, a thruster-only actuation system is used for the 6-DOF pose control. Within this setting, the control force $\mathbf{f}$ and control torque $\boldsymbol{\tau}$ are given by

$$[\mathbf{f}^T, \boldsymbol{\tau}^T]^T = \mathbf{A}\mathbf{u} \quad (14)$$

where $\mathbf{A} \in \mathbb{R}^{6 \times N}$ is thrust allocation matrix, $\mathbf{u} \in \mathbb{R}^N$ is the output vector of N thruster pairs, and N denotes the number of thruster pairs. As shown in Fig. 3, the servicing spacecraft is equipped with 12 thrusters, which are arranged into six thruster pairs, denoted by $\{T_i, L_i\}$, $i = 1, 2, \dots, 6$. Each thruster pair can provide bidirectional thrusts, and the allocation matrix can be written as [33]

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & -d_z & d_z & 0 & 0 \\ 0 & 0 & 0 & 0 & -d_x & d_x \\ -d_y & d_y & 0 & 0 & 0 & 0 \end{bmatrix} \quad (15)$$

where d_x , d_y , and d_z represent the moment arms of thrusts with respect to the center of mass of the servicing spacecraft along three axes of $\mathcal{S}$. In addition, each thruster has a limited output in practical engineering, that is,

$$|u_i| \leq \gamma, i = 1, 2, \dots, 6 \quad (16)$$

where γ is the maximum value of thrust.

D. Problem Formulation

Define $\mathbf{x}_1 = [\boldsymbol{\rho}_e^T, \mathbf{q}_{ev}^T]^T$ and $\mathbf{x}_2 = [(\mathbf{M}\mathbf{v}_e)^T, (\mathbf{J}_s\boldsymbol{\omega}_e)^T]^T$, then based on (6), (10), and (11), the integrated 6-DOF

relative motion dynamics can be written as

$$\dot{x}_1 = \begin{bmatrix} M^{-1} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & BJ_s^{-1} \end{bmatrix} x_2 \quad (17)$$

$$\dot{x}_2 = l + Au + b \quad (18)$$

where $B = 0.5[S(q_{ev}) + q_{e4}I_3]$, $l = [h_p^\top, h_a^\top]^\top$ with $h_p = -Cv_e - G$, $h_a = -S(\omega_{SN}^S)J_s\omega_{SN}^S + J_s(S(\omega_e)R_{ST}\omega_{TN}^\tau - R_{ST}\omega_{TN}^\tau)$, and $b = [f_d^\top, \tau_d^\top]^\top$ is the disturbance vector. Although the relative position and attitude motions are modeled separately for flexibility, the integrated dynamics (18) is, in essence, a coupled 6-DOF relative motion dynamics. The couplings mainly arise from three aspects: inherent dynamics, environmental disturbances, and thruster configuration.

- 1) It is clear from (6) that the relative position dynamics is affected by the attitude motion. Specifically, the matrices C and D show the mutual coupling between position and attitude motions caused by the rotation of servicing spacecraft (i.e., ω_{SN}^S). Similar coupling terms can also be found in [35], where the relative position dynamics was established in $\mathcal{L}$, and matrices C and D are related to the orbital angular velocity of the target. Furthermore, the vector w_2 in nonlinear term G involves the rotation matrix R_{ST} (due to the relation $\rho_d = R_{ST}p_d$) and relative angular velocity ω_e , implying mutual couplings induced by relative attitude information.
- 2) External disturbances also cause position and attitude mutual couplings [35], [36]. The J_2 perturbing force $f_d^{J_2}$ is related to the servicing spacecraft attitude R_{SN} , whereas the gravity-gradient disturbance torque τ_d^g depends on the position vector ρ_s , as shown in (12).
- 3) A thruster-only actuation system, as shown in Fig. 3, is adopted for 6-DOF pose control, which further induces the position and attitude mutual coupling.

REMARK 1 The dual quaternion-based modeling method has received wide attention in the literature (e.g., see [32] and [37]), due to its ability to simultaneously describe the position and attitude motions of spacecraft while inherently capturing their mutual couplings. However, recent studies have identified certain deficiencies in this modeling method, specifically related to physical clearance, information matching, and expansibility [35]. In contrast, the separated-integrated relative motion dynamics, as described by (17) and (18), not only accounts for the mutual couplings between the position and attitude motions, but also offers more flexibility for control design. In fact, the dynamics derived under the dual algebraic framework can be decomposed into real and dual parts, and the decomposed subdynamics are consistent with the separated-integrated modeling method. It is noted that integrating specific actuator configurations and physical limitations into the dual quaternion-based modeling framework is cumbersome due to the representation of control inputs in dual form. In contrast, the separated-integrated modeling method explicitly

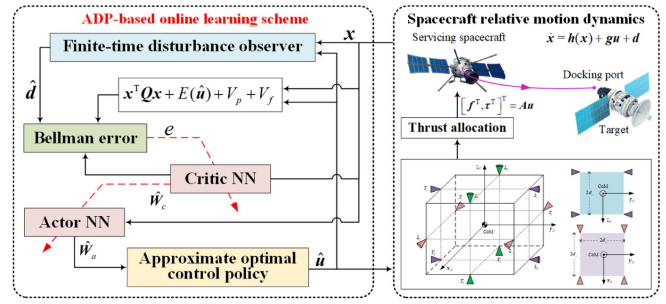


Fig. 4. Block diagram of the proposed control scheme.

describes the mapping between control inputs and actuator outputs [see (14)], providing a more convenient way to address the limited thrust problem.

Control objective: Consider the spacecraft relative position and attitude dynamics described by (17), and (18). Design an optimal saturated control scheme to achieve

$$\lim_{t \rightarrow \infty} \text{col}(x_1(t), x_2(t)) = \mathbf{0}$$

while ensuring online performance optimization and the satisfaction of motion and input constraints, in the presence of external disturbances.

III. MAIN RESULTS

A finite-time disturbance observer is developed to provide an accurate estimation of external disturbances. By doing so, the ADP-based online learning algorithm can be updated in resistance to environmental disturbances. To cope with motion and input constraints, an especially designed performance index is constructed for the optimal control problem by introducing two barrier functions and a nonquadratic term. By employing the ADP technique, the approximation to optimal control policy is learned online. The block diagram of the proposed control scheme is shown in Fig. 4.

A. Finite-Time Disturbance Observer

Inspired by [38], we define an auxiliary variable $s = x_2 - z$ with z given by

$$\begin{aligned} \dot{z} = & ks + \chi[s] + \varepsilon[s]\|s\|^{m/n} \\ & + \|I\|s] + Au \end{aligned} \quad (19)$$

where m and n ($m < n$) are odd positive integers; k , ε , and χ ($\chi \geq \|b\|$) are positive constants. In addition, $[\cdot]$ is a norm-normalized sign function, defined by

$$[x] = \begin{cases} \frac{x}{\|x\|}, & \text{if } x \neq \mathbf{0} \\ \mathbf{0}, & \text{if } x = \mathbf{0}. \end{cases} \quad (20)$$

According to the idea of terminal sliding mode, the finite-time disturbance observer is designed as

$$\begin{aligned} \dot{\hat{b}} = & ks + \chi[s] + \varepsilon[s]\|s\|^{m/n} \\ & + \|I\|s] - I. \end{aligned} \quad (21)$$

LEMMA 1 Consider the perturbed system as in (18). The DOB designed in (21) can ensure that the disturbance estimation error $\tilde{\mathbf{b}} \triangleq \mathbf{b} - \hat{\mathbf{b}}$ converges to zero in a finite time.

PROOF 1 Taking time derivative of s yields $\dot{s} = \dot{\mathbf{x}}_2 - \dot{\mathbf{z}} = \tilde{\mathbf{b}}$. Then, we consider the Lyapunov candidate $V_1 = 0.5s^\top s$, whose time derivative is evaluated as

$$\begin{aligned}\dot{V}_1 &= -ks^\top s - \chi\|s\| - \varepsilon\|s\|^{(m+n)/n} \\ &\quad - \|l\|\|s\| + s^\top l + s^\top b \\ &\leq -ks^\top s - \varepsilon\|s\|^{(m+n)/n} \\ &= -2kV_1 - 2^{(m+n)/2n}\varepsilon V_1^{(m+n)/2n}.\end{aligned}\quad (22)$$

Hence, the sliding variable s converges to origin in a finite time [39], indicating that the disturbance estimation error $\tilde{\mathbf{b}}$ converges to zero in a finite time. ■

B. Learning-Based Control Scheme

Rewrite the integrated pose tracking dynamics of spacecraft into the following compact form:

$$\dot{\mathbf{x}} = \mathbf{h}(\mathbf{x}) + \mathbf{g}\mathbf{u} + \mathbf{d} \quad (23)$$

where $\mathbf{x} = [\mathbf{x}_1^\top, \mathbf{x}_2^\top]^\top$, and

$$\mathbf{h}(\mathbf{x}) = \begin{bmatrix} v_e \\ \mathbf{B}\boldsymbol{\omega}_e \\ \mathbf{h}_p \\ \mathbf{h}_a \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} \mathbf{0}_{6 \times 6} \\ \mathbf{A} \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} \mathbf{0}_{6 \times 1} \\ \mathbf{b} \end{bmatrix}. \quad (24)$$

To proceed, the following assumptions are made.

ASSUMPTION 1 The disturbance $\mathbf{d}$ is unknown but bounded, that is, $\|\mathbf{d}\| \leq b_d$ for some positive constant b_d .

ASSUMPTION 2 The function $\mathbf{h}(\mathbf{x})$ is Lipschitz continuous, that is, $\|\mathbf{h}(\mathbf{x})\| \leq b_h\|\mathbf{x}\|$ with a Lipschitz constant $b_h > 0$. In addition, $\mathbf{g}$ is bounded by $\|\mathbf{g}\| < b_g$.

The performance index function for the optimal control is constructed as follows:

$$V(\mathbf{x}(t)) = \int_t^\infty T(\mathbf{x}(\sigma), \mathbf{u}(\sigma))d\sigma \quad (25)$$

where $T(\mathbf{x}, \mathbf{u}) = \mathbf{x}^\top \mathbf{Q}\mathbf{x} + E(\mathbf{u}) + V_p + V_f$, and $E(\mathbf{u})$ is designed as the following nonquadratic form to address input constraints [40]

$$E(\mathbf{u}) = 2\gamma \int_0^u \tanh^{-1}(\sigma/\gamma)d\sigma. \quad (26)$$

To handle the path constraint, V_p is formulated as

$$V_p = \frac{k_1 \boldsymbol{\rho}_e^\top \boldsymbol{\rho}_e}{(\boldsymbol{\rho}/\|\boldsymbol{\rho}\|) \cdot \mathbf{R}_{ST} \cdot \mathbf{n}_1 - \cos \alpha_m} \quad (27)$$

where $k_1 > 0$ is a tunable parameter. It is noted that the nonuniqueness of quaternion-based attitude representation may result in the unwinding phenomenon, which would become evident when considering a tumbling target. Toward this end, an antiunwinding attitude error term is developed

to avoid the unwinding problem, which is formulated as

$$\Theta = \frac{1 - q_{e4}^2}{q_{e4}^2}. \quad (28)$$

Considering both the unwinding problem and FOV constraint, the barrier function V_f is then designed as

$$V_f = \frac{k_2 \Theta}{(-\boldsymbol{\rho}/\|\boldsymbol{\rho}\|) \cdot \mathbf{n}_2 - \cos \beta_m} \quad (29)$$

where $k_2 > 0$ is a tunable parameter.

REMARK 2 Discontinuous control [41], [42] usually employs the sign function to generate discontinuous feedback, which can avoid the unwinding phenomenon. However, it is not robust to small measurement noise and may cause the chattering phenomenon. In the hybrid control methods, such as [43] and [44], a logic variable is utilized to specify the desired orientation of rotation, which alleviates the undesirable results of unwinding and achieves global attitude stabilization. However, it is rather difficult to incorporate these approaches into the ADP-based control framework. In this article, an antiunwinding attitude error function Θ is introduced into V_f to avoid the unwinding phenomenon. Then, from the boundedness of V_f (as will be shown in Theorem 1), it follows that the error quaternion $\mathbf{q}_e$ evolves within the set $\mathbb{C}_q = \{\mathbf{q}_e \in \mathbb{S}^3 \mid q_{e4} \neq 0\}$. In fact, the set $\mathbb{C}_q$ divides the unit sphere $\mathbb{S}^3$ into two disjoint sections: one section corresponds to the subset where $q_{e4} > 0$, while the other one corresponds to the subset where $q_{e4} < 0$. As such, if $q_{e4}(t) \neq 0$ is guaranteed for all $t \geq 0$, then the sign of q_{e4} will remain unchanged and thus the unwinding phenomenon is avoided. One caveat is that this requires $q_{e4}(0) \neq 0$.

Define the Hamiltonian of this problem as

$$H(\mathbf{x}, \mathbf{u}, \nabla V) = T(\mathbf{x}, \mathbf{u}) + \nabla V^\top (\mathbf{h} + \mathbf{g}\mathbf{u} + \mathbf{d}) \quad (30)$$

where $\nabla(\cdot)$ is the partial derivative $\partial(\cdot)/\partial \mathbf{x}$. The optimal performance index function $V^*(\mathbf{x})$ satisfies the Hamilton–Jacobi–Bellman (HJB) equation

$$\begin{aligned}\min_{\mathbf{u}} H(\mathbf{x}, \mathbf{u}, \nabla V^*) \\ = \min_{\mathbf{u}} [T + (\nabla V^*)^\top (\mathbf{h} + \mathbf{g}\mathbf{u} + \mathbf{d})] = 0.\end{aligned}\quad (31)$$

Differentiating both sides of (31) with respect to $\mathbf{u}$, we get the optimal control policy as

$$\mathbf{u}^*(\mathbf{x}) = -\gamma \tanh(\boldsymbol{\xi}^*) \quad (32)$$

where $\boldsymbol{\xi}^* = (1/2\gamma)\mathbf{g}^\top \nabla V^*$. Recalling the properties of the hyperbolic function, (26) becomes

$$\begin{aligned}E(\mathbf{u}^*) &= \gamma \mathbf{g}^\top \nabla V^* \tanh(\boldsymbol{\xi}^*) \\ &\quad + \gamma^2 \sum_{i=1}^3 \ln(1 - \tanh^2(\xi_i^*)).\end{aligned}\quad (33)$$

By substituting the optimal control (32) into the Hamiltonian (30), the HJB equation in terms of ∇V^* can be

rewritten as

$$\begin{aligned} & \mathbf{x}^\top \mathbf{Q} \mathbf{x} + V_p + V_f + (\nabla V^*)^\top (\mathbf{h} + \mathbf{d}) \\ & + \gamma^2 \sum_{i=1}^3 \ln(1 - \tanh^2(\xi_i^*)) = 0 \end{aligned} \quad (34)$$

with $V^*(\mathbf{0}) = 0$. It is obvious that the optimal control policy can be obtained by solving the HJB equation. However, since the HJB equation is a nonlinear partial differential equation that is hard to solve, it is difficult or even impossible to get the solution of the above HJB equation in an analytical way. To overcome this problem, the ADP technique is introduced, which uses online learning NNs to estimate the performance index function and derive a near-optimal solution.

According to the NN approximation theorem, the optimal performance index and control policy can be approximated by NNs as

$$V^*(\mathbf{x}) = \mathbf{W}^\top \boldsymbol{\sigma}(\mathbf{x}) + \varepsilon(\mathbf{x}) \quad (35)$$

$$\mathbf{u}^*(\mathbf{x}) = -\gamma \tanh \left(\frac{1}{2\gamma} \mathbf{g}^\top (\nabla \boldsymbol{\sigma}^\top \mathbf{W} + \nabla \varepsilon^\top) \right) \quad (36)$$

where $\mathbf{W} \in \mathbb{R}^m$, $\boldsymbol{\sigma} \in \mathbb{R}^m$ (m is the number of neurons), and ε are the network weight, basis function, and approximation error, respectively. Two NNs are designed to approximate the optimal performance index and control policy, which can be represented as critic $\hat{V}(\mathbf{x})$ and actor $\hat{\mathbf{u}}(\mathbf{x})$, given by

$$\hat{V}(\mathbf{x}) = \hat{\mathbf{W}}_c^\top \boldsymbol{\sigma}(\mathbf{x}) \quad (37)$$

$$\hat{\mathbf{u}}(\mathbf{x}) = -\gamma \tanh \left(\frac{1}{2\gamma} \mathbf{g}^\top \nabla \boldsymbol{\sigma}^\top \hat{\mathbf{W}}_a \right) \quad (38)$$

where $\hat{\mathbf{W}}_c \in \mathbb{R}^m$ and $\hat{\mathbf{W}}_a \in \mathbb{R}^m$ are the estimated values of the ideal weights of the critic and actor NNs, respectively. This kind of NN estimation can be regarded as the polynomial approximation [17], which has been widely used in the ADP-based near-optimal control schemes. According to the Weierstrass approximation theorem, there exists a completely independent basis set $\{\sigma_i(\mathbf{x})\}$ such that the cost function $V^*(\mathbf{x})$ is uniformly approximated. When compared with the ‘‘Sigmoid,’’ ‘‘ReLU,’’ or other activation functions, this polynomial-type function can easily construct a state-feedback control policy [25]. That is, by selecting state-dependent basis functions and specifying the network weights as coefficients of the controller, the defined NNs can generate some commonly used state-feedback controllers. Let us define $\hat{\xi} = (1/2\gamma) \mathbf{g}^\top \nabla \boldsymbol{\sigma}^\top \hat{\mathbf{W}}_a$ for notational brevity, and define $\tilde{\mathbf{W}}_c = \mathbf{W} - \hat{\mathbf{W}}_c$ and $\tilde{\mathbf{W}}_a = \mathbf{W} - \hat{\mathbf{W}}_a$ as the estimation error of the critic and actor NNs, respectively. The following basic assumption that is necessary for the ADP-based control design is made [45].

ASSUMPTION 3 For a permissible control policy, there exist positive constants b_σ , b_ε , $b_{\sigma x}$, and $b_{\varepsilon x}$ such that $\|\boldsymbol{\sigma}\| \leq b_\sigma$, $|\varepsilon| \leq b_\varepsilon$, $\|\nabla \boldsymbol{\sigma}\| \leq b_{\sigma x}$, and $\|\nabla \varepsilon\| \leq b_{\varepsilon x}$.

The approximated Hamiltonian is written as

$$\begin{aligned} e &= \hat{H}(\mathbf{x}, \hat{\mathbf{u}}, \nabla \hat{V}) \\ &= T(\mathbf{x}, \hat{\mathbf{u}}) + \hat{\mathbf{W}}_c^\top (\mathbf{h} + \mathbf{g}\hat{\mathbf{u}} + \hat{\mathbf{d}}). \end{aligned} \quad (39)$$

Then, the Bellman residual error between the optimal Hamiltonian and the estimated one is

$$\begin{aligned} e &= \hat{H}(\mathbf{x}, \hat{\mathbf{u}}, \nabla \hat{V}) - H(\mathbf{x}, \mathbf{u}^*, \nabla V^*) \\ &= -\tilde{\mathbf{W}}_c^\top \boldsymbol{\phi} - \mathbf{W}^\top \nabla \sigma \tilde{\mathbf{d}} + \mathbf{W}^\top \nabla \sigma \mathbf{g} \gamma \\ &\quad \times (\tanh(\hat{\xi}^*) - \tanh(\hat{\xi})) + E(\hat{\mathbf{u}}) - E(\mathbf{u}^*) \\ &\quad - \nabla \varepsilon^\top (\mathbf{h} + \mathbf{g}\mathbf{u}^* + \mathbf{d}) \end{aligned} \quad (40)$$

where $\boldsymbol{\phi} = \nabla \sigma(\mathbf{h} + \mathbf{g}\hat{\mathbf{u}} + \hat{\mathbf{d}})$, $\hat{\mathbf{d}} = [\mathbf{0}_{6 \times 1}^\top, \hat{\mathbf{b}}^\top]^\top$ is the estimation of $\mathbf{d}$, and $\tilde{\mathbf{d}} = \mathbf{d} - \hat{\mathbf{d}}$ is the estimation error. For notational brevity, we further define $\boldsymbol{\phi}_1 = \boldsymbol{\phi}/(1 + \boldsymbol{\phi}^\top \boldsymbol{\phi})$ and $\boldsymbol{\phi}_2 = \boldsymbol{\phi}_1/(1 + \boldsymbol{\phi}^\top \boldsymbol{\phi})$. The critic NN weights should be tuned to minimize $\Delta = 0.5e^2$. Thus, using the normalized gradient descent algorithm, we have

$$\begin{aligned} \dot{\hat{\mathbf{W}}}_c &= -\frac{\eta_1}{(1 + \boldsymbol{\phi}^\top \boldsymbol{\phi})^2} \frac{\partial \Delta}{\partial \hat{\mathbf{W}}_c} - \eta_2 \boldsymbol{\Phi} \\ &= -\eta_1 \boldsymbol{\phi}_2 e - \eta_2 \boldsymbol{\Phi}, \end{aligned} \quad (41)$$

where η_1 and η_2 are positive constants, and $\boldsymbol{\Phi}$ is introduced to relax the signal excitation condition for convergence of critic NN weights [24]

$$\boldsymbol{\Phi}(t, t_1, t_s) = \delta_1(t_1, t_s) \hat{\mathbf{W}}_c(t) + \delta_2(t_1, t_s) \quad (42)$$

with

$$\dot{\delta}_1(t, t_1) = -\kappa \delta_1 + \boldsymbol{\phi}_1^\top \boldsymbol{\phi}_1, \quad \dot{\delta}_2(t, t_1) = -\kappa \delta_2 + T \boldsymbol{\phi}_2 \quad (43)$$

where $\kappa > 0$, $\delta_1(t_1) = \mathbf{0}_{m \times m}$, and $\delta_2(t_1) = \mathbf{0}_{m \times 1}$. Substituting the solution of (43) into (42) yields

$$\begin{aligned} \boldsymbol{\Phi}(t, t_1, t_s) &= \int_0^{t_s} e^{\kappa(\tau - t_s)} (\boldsymbol{\phi}_1(\tau) \boldsymbol{\phi}_1^\top(\tau) \hat{\mathbf{W}}_c(t) + T \boldsymbol{\phi}_2(\tau)) d\tau \\ &= -\delta_1(t_1, t_s) \tilde{\mathbf{W}}_c(t) + \boldsymbol{\varepsilon}_\Phi \end{aligned} \quad (44)$$

where $\delta_1(t_1, t_s) = \int_{t_1}^{t_s} e^{\kappa(\tau - t_s)} \boldsymbol{\phi}_1(\tau) \boldsymbol{\phi}_1^\top(\tau) d\tau$ is regarded as an information matrix that records the historical data of $\boldsymbol{\phi}$ on time interval $[t_1, t_s]$, and $\boldsymbol{\varepsilon}_\Phi$ is a bounded residual error. Under the assumption that $\boldsymbol{\phi}$ satisfies a finite excitation (FE) condition, $\delta_1 \geq c_\Phi \mathbf{I}_m$ holds with a positive constant c_Φ [24]. In addition, the actor NN is updated as

$$\dot{\hat{\mathbf{W}}}_a = -\eta_3 (\mathbf{P}_1 \hat{\mathbf{W}}_a + \nabla \sigma \mathbf{g} \gamma \tanh(\hat{\xi}) - \mathbf{P}_2 \hat{\mathbf{W}}_c) \quad (45)$$

where $\eta_3 > 0$, and $\mathbf{P}_1$ and $\mathbf{P}_2$ are positive definite matrices.

REMARK 3 In practical implementation, the update laws (41) and (45) can be solved employing the fixed-step Runge–Kutta method with a small sample time Δt . Other methods, such as zero-order holder (ZOH), can also be used to get the numerical solutions at each sampling step, which yields $\hat{\mathbf{W}}_c(t + \Delta t) = \hat{\mathbf{W}}_c(t) + \dot{\hat{\mathbf{W}}}_c(t) \Delta t$, and $\hat{\mathbf{W}}_a(t + \Delta t) = \hat{\mathbf{W}}_a(t) + \dot{\hat{\mathbf{W}}}_a(t) \Delta t$, ($t = 0, \Delta t, \dots, T$). For an operation with a duration of T , the total number of update steps is equal to $T/\Delta t$. In addition, inspecting (41) and

(45) reveals that $\dot{\mathbf{W}}_c(t)$ depends on ϕ_2 , e , and Φ , while $\dot{\mathbf{W}}_a(t)$ is calculated based on $\hat{\mathbf{W}}_a$, $\hat{\mathbf{W}}_c$, $\nabla\sigma$, and $\mathbf{g}$. It is easy to check that all variables used to calculate $\dot{\mathbf{W}}_c(t)$ and $\dot{\mathbf{W}}_a(t)$ can be obtained based on measurable signals. Thus, the two update laws can be easily implemented in practice.

C. Stability Analysis

THEOREM 1 Consider the spacecraft integrated pose tracking dynamics (23) with initial states satisfying (7) and (13). The saturated controller (38) together with the NN update laws (41) and (45) can ensure that the system states and network weight estimation errors are uniformly ultimately bounded (UUB), and that path and FOV constraints are never violated.

PROOF 2 Choose an overall Lyapunov candidate as $L = V^* + L_1 + L_2 + V_1$, with $L_1 = (1/2\eta_1)\tilde{\mathbf{W}}_c^\top \tilde{\mathbf{W}}_c$ and $L_2 = (1/2\eta_3)\tilde{\mathbf{W}}_a^\top \tilde{\mathbf{W}}_a$. The time derivative of V^* is evaluated as

$$\begin{aligned}\dot{V}^* &= (\mathbf{W}^\top \nabla \sigma + \nabla \varepsilon^\top)(\mathbf{h} + \mathbf{g}\hat{\mathbf{u}} + \mathbf{d}) \\ &= \mathbf{W}^\top \nabla \sigma(\mathbf{h} + \mathbf{d}) - \mathbf{W}^\top \nabla \sigma \gamma \tanh(\hat{\xi}) \\ &\quad + \nabla \varepsilon^\top(\mathbf{h} + \mathbf{g}\hat{\mathbf{u}} + \mathbf{d}) \\ &\leq \mathbf{W}^\top \nabla \sigma(\mathbf{h} + \mathbf{d}) - \mathbf{W}^\top \nabla \sigma \gamma \tanh(\hat{\xi}) \\ &\quad + b_{\varepsilon x} b_h \|\mathbf{x}\| + \varepsilon_1\end{aligned}\quad (46)$$

where $\varepsilon_1 = b_{\varepsilon x}(\gamma b_g + b_d)$. If the optimal performance index function is approximated with no reconstruction error, the HJB equation can be written as

$$\begin{aligned}H(\mathbf{x}, \mathbf{u}, \nabla V) &= \mathbf{x}^\top \mathbf{Q} \mathbf{x} + V_p + V_f + \mathbf{W}^\top \nabla \sigma(\mathbf{h} + \mathbf{d}) \\ &\quad + \gamma^2 \sum_{i=1}^3 \ln(1 - \tanh^2(\xi_i))\end{aligned}\quad (47)$$

where $\xi = (1/2\gamma)\mathbf{g}^\top \nabla \sigma^\top \mathbf{W}$, $\mathbf{u} = -\gamma \tanh(\xi)$, and $E(\mathbf{u}) = \mathbf{W}^\top \nabla \sigma \gamma \tanh(\xi) + \gamma^2 \sum_{i=1}^3 \ln(1 - \tanh^2(\xi_i))$ is defined for brevity. Then, the approximation error of the HJB equation can be given by

$$\begin{aligned}\varepsilon_{\text{HJB}} &= H(\mathbf{x}, \mathbf{u}^*, \nabla V^*) - H(\mathbf{x}, \mathbf{u}, \nabla V) \\ &= \gamma^2 \sum_{i=1}^3 [\ln(1 - \tanh^2(\xi_i^*)) \\ &\quad - \ln(1 - \tanh^2(\xi_i))] + \nabla \varepsilon^\top(\mathbf{h} + \mathbf{d})\end{aligned}\quad (48)$$

where $|\varepsilon_{\text{HJB}}| \leq b_{\varepsilon x} b_h \|\mathbf{x}\| + \varepsilon_2$, with $\varepsilon_2 = \gamma^2 \sum_{i=1}^3 |\ln(1 - \tanh^2(\xi_i^*)) - \ln(1 - \tanh^2(\xi_i))| + b_{\varepsilon x} b_d$. Then, one has

$$\begin{aligned}\mathbf{x}^\top \mathbf{Q} \mathbf{x} + V_p + V_f + \mathbf{W}^\top \nabla \sigma(\mathbf{h} + \mathbf{d}) \\ + \gamma^2 \sum_{i=1}^3 \ln(1 - \tanh^2(\xi_i)) + \varepsilon_{\text{HJB}} = 0.\end{aligned}\quad (49)$$

Using the aforementioned equation in (46), one gets

$$\begin{aligned}\dot{V}^* &\leq -\mathbf{x}^\top \mathbf{Q} \mathbf{x} - V_p - V_f - \gamma^2 \sum_{i=1}^3 \ln(1 - \tanh^2(\xi_i)) \\ &\quad - \varepsilon_{\text{HJB}} - \mathbf{W}^\top \nabla \sigma \gamma \tanh(\hat{\xi}) + b_{\varepsilon x} b_h \|\mathbf{x}\| + \varepsilon_1 \\ &= -\mathbf{x}^\top \mathbf{Q} \mathbf{x} - V_p - V_f + \mathbf{W}^\top \nabla \sigma \gamma [\tanh(\xi) \\ &\quad - \tanh(\hat{\xi})] - E(\mathbf{u}) - \varepsilon_{\text{HJB}} + b_{\varepsilon x} b_h \|\mathbf{x}\| + \varepsilon_1 \\ &\leq -\mathbf{x}^\top \mathbf{Q} \mathbf{x} - V_p - V_f - E(\mathbf{u}) + k_x \|\mathbf{x}\| + \varepsilon_3 \\ &\quad + \mathbf{W}^\top \nabla \sigma \gamma \tanh(\xi) - \tilde{\mathbf{W}}_a^\top \nabla \sigma \gamma \tanh(\hat{\xi})\end{aligned}\quad (50)$$

where $\tilde{\mathbf{W}}_a^\top \nabla \sigma \gamma \tanh(\hat{\xi}) \geq 0$ is used, $k_x = 2b_{\varepsilon x} b_h$, and $\varepsilon_3 = \varepsilon_1 + \varepsilon_2$. Differentiating L_1 and using (41) gives

$$\dot{L}_1 = \tilde{\mathbf{W}}_c^\top \phi_2 e + (\eta_2/\eta_1)\tilde{\mathbf{W}}_c^\top \Phi. \quad (51)$$

From (40) and (48), it yields that

$$\begin{aligned}e &= \hat{H}(\mathbf{x}, \hat{\mathbf{u}}, \nabla \hat{V}) - H(\mathbf{x}, \mathbf{u}, \nabla V) - \varepsilon_{\text{HJB}} \\ &= E(\hat{\mathbf{u}}) - E(\mathbf{u}) + \tilde{\mathbf{W}}_c^\top \nabla \sigma(\mathbf{h} + \mathbf{g}\hat{\mathbf{u}} + \hat{\mathbf{d}}) \\ &\quad - \mathbf{W}^\top \nabla \sigma(\mathbf{h} + \mathbf{g}\mathbf{u} + \mathbf{d}) - \varepsilon_{\text{HJB}} \\ &= E(\hat{\mathbf{u}}) - E(\mathbf{u}) - \tilde{\mathbf{W}}_c^\top \phi + \mathbf{W}^\top \nabla \sigma \gamma [\tanh(\xi) \\ &\quad - \tanh(\hat{\xi})] - \mathbf{W}^\top \nabla \sigma \tilde{\mathbf{d}} - \varepsilon_{\text{HJB}}.\end{aligned}\quad (52)$$

Considering the first term in the right-hand side of the aforementioned equation, we have

$$\begin{aligned}E(\hat{\mathbf{u}}) - E(\mathbf{u}) &= \tilde{\mathbf{W}}_a^\top \nabla \sigma \gamma \tanh(\hat{\xi}) + \gamma^2 \sum_{i=1}^3 \ln(1 - \tanh^2(\hat{\xi}_i)) \\ &\quad - \mathbf{W}^\top \nabla \sigma \gamma \tanh(\xi) - \gamma^2 \sum_{i=1}^3 \ln(1 - \tanh^2(\xi_i)).\end{aligned}\quad (53)$$

Using the property $\ln(1 - \tanh^2(x)) = \ln(4) - 2x \tanh(x) + \varepsilon_x$ with a bounded approximation error ε_x , it gets

$$E(\hat{\mathbf{u}}) - E(\mathbf{u}) = \gamma^2(\varepsilon_{\hat{\xi}} - \varepsilon_{\xi}). \quad (54)$$

We further have

$$\begin{aligned}\dot{L}_1 &= \tilde{\mathbf{W}}_c^\top \phi_2 [\gamma^2(\varepsilon_{\hat{\xi}} - \varepsilon_{\xi}) - \tilde{\mathbf{W}}_c^\top \phi \\ &\quad + \mathbf{W}^\top \nabla \sigma \gamma [\tanh(\xi) - \tanh(\hat{\xi})] \\ &\quad - \mathbf{W}^\top \nabla \sigma \tilde{\mathbf{d}} - \varepsilon_{\text{HJB}}] + (\eta_2/\eta_1)\tilde{\mathbf{W}}_c^\top \Phi \\ &\leq -\tilde{\mathbf{W}}_c^\top [(\eta_2/\eta_1)c_\Phi \mathbf{I} + \phi_1 \phi_1^\top - 0.5b_{\varepsilon x} b_h] \tilde{\mathbf{W}}_c \\ &\quad + k_c \|\tilde{\mathbf{W}}_c\| + 0.5b_{\varepsilon x} b_h \|\mathbf{x}\|^2\end{aligned}\quad (55)$$

where $k_c = \|\phi_2\|[\gamma^2(\varepsilon_{\hat{\xi}} - \varepsilon_{\xi}) + b_{\varepsilon x} \|\mathbf{W}\|(2b_g \gamma + \|\tilde{\mathbf{d}}\|) + \varepsilon_2] + (\eta_2/\eta_1)\varepsilon_\Phi$, with $\tilde{\mathbf{d}}$ converging to zero in the finite time. Finally, using (45) and recalling that $\dot{V}_1 \leq 0$, the time derivative of L becomes

$$\begin{aligned}\dot{L} &\leq -\mathbf{x}^\top \mathbf{H} \mathbf{x} - V_p - V_f - E(\mathbf{u}) + k_x \|\mathbf{x}\| + k_L \\ &\quad - \tilde{\mathbf{W}}_c^\top \mathbf{Y} \tilde{\mathbf{W}}_c + k_c \|\tilde{\mathbf{W}}_c\| + k_a \|\tilde{\mathbf{W}}_a\| \\ &\quad - \tilde{\mathbf{W}}_a^\top \mathbf{Z} \tilde{\mathbf{W}}_a\end{aligned}\quad (56)$$

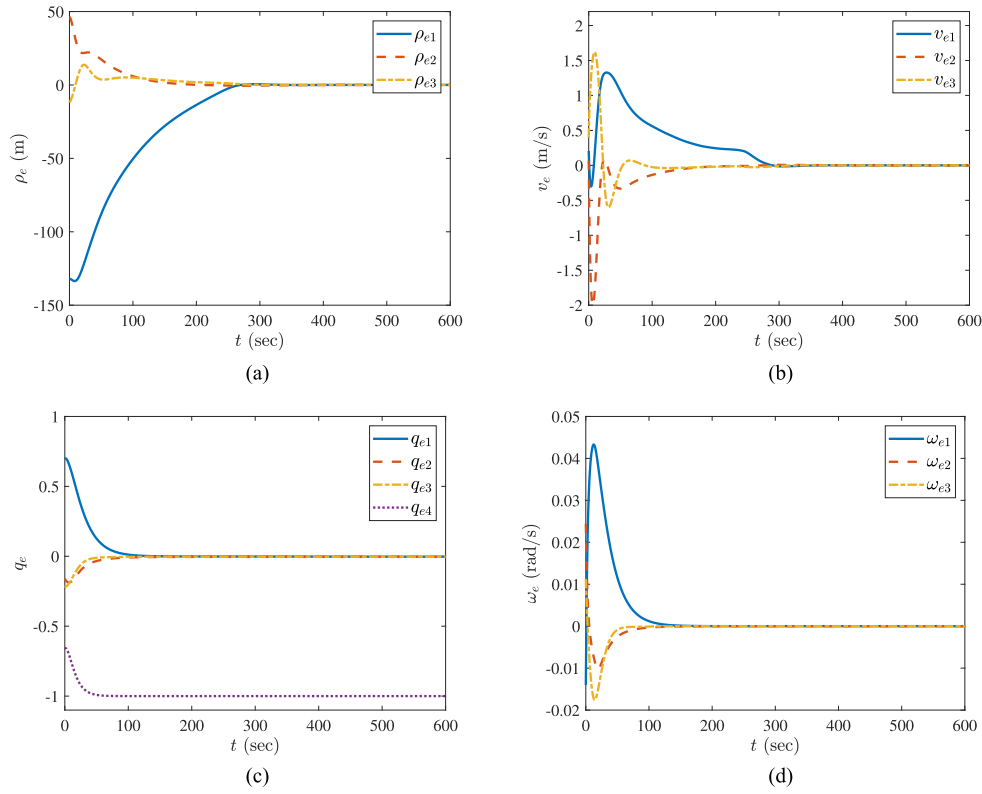


Fig. 5. Time responses of the relative pose tracking. (a) Relative position errors. (b) Relative velocity errors. (c) Attitude errors. (d) Angular rate errors.

where $\mathbf{H} = \mathbf{Q} - 0.5b_{\varepsilon x}b_h\mathbf{I}$, $\mathbf{Y} = (\eta_2/\eta_1)c_\Phi\mathbf{I} + \boldsymbol{\phi}_1\boldsymbol{\phi}_1^\top - 0.5(b_{\varepsilon x}b_h + \eta_4)$, $\mathbf{Z} = \mathbf{P} - 0.5\eta_4\mathbf{I}$, $k_L = b_{\varepsilon x}b_g\|\mathbf{W}\| + \varepsilon_3$, and $k_a = \|\mathbf{P}\mathbf{W}\| + \|\mathbf{W}\|$. Since the performance index remains bounded under a permissible control, $\mathbf{W}$ is regarded as a bounded weight, which indicates that k_L , k_c , and k_a are bounded. Therefore, by adjusting $\lambda_m(\mathbf{Q}) > 0.5b_{\varepsilon x}b_h$, $(\eta_2/\eta_1)c_\Phi > 0.5(b_{\varepsilon x}b_h + \eta_4)$, $\lambda_m(\mathbf{P}) > 0.5\eta_4$ and further defining $\mathbf{y} = [\mathbf{x}^\top, \tilde{\mathbf{W}}_c^\top, \tilde{\mathbf{W}}_a^\top]^\top$, it is concluded that $\dot{L} \leq 0$ if

$$\|\mathbf{y}\| > \frac{k_y}{2\lambda_m(\mathbf{\Gamma})} + \sqrt{\frac{k_y^2}{4\lambda_m^2(\mathbf{\Gamma})} + \frac{k_L}{\lambda_m(\mathbf{\Gamma})}} \quad (57)$$

where $\mathbf{\Gamma} = \text{blkdiag}[\mathbf{H}, \mathbf{Y}, \mathbf{Z}]$, and $k_y = k_x + k_c + k_a$. This indicates that $\mathbf{x}$, $\tilde{\mathbf{W}}_c$, and $\tilde{\mathbf{W}}_a$ are UUB, and that $V_p, V_f \in \mathcal{L}_\infty$. From the latter, it is clear that neither path nor FOV constraints are ever violated. This completes the proof. ■

REMARK 4 The DOB technique is incorporated into the online learning scheme to enhance the antidisturbance ability of the model-based ADP algorithm. Specifically, the estimation of overall disturbances is integrated into signal $\boldsymbol{\phi} = \nabla\sigma(\mathbf{h} + \mathbf{g}\mathbf{u} + \hat{\mathbf{d}})$, which is further used for updating the critic NN. Benefiting from the finite-time convergence property of the DOB (21), the proposed online learning scheme is tuned by an accurate model dynamics, despite the presence of external disturbances.

REMARK 5 The barrier functions V_p and V_f are introduced in the performance index function V to deal with path and FOV constraints. It is clear that V_p and V_f become zero

when the pose tracking errors $\boldsymbol{\rho}_e$ and $\mathbf{q}_{ev}$ converge to zero. In addition, based on (7) and (13), it can be seen that $V_p, V_f \rightarrow \infty$ at the boundaries of motion constraints. Therefore, if the initial conditions satisfy (7) and (13), then we can guarantee that the servicing spacecraft complies with path and FOV constraints by ensuring the boundedness of $V(\mathbf{x})$.

REMARK 6 The convergence of the critic NN usually requires persistent excitation (PE) condition of $\boldsymbol{\phi}$ [46], which is rarely met in practical proximity operations. In this article, Φ is introduced into (41) to relax the restrictive PE condition to a strictly weaker FE condition.

REMARK 7 Designing a proper initial policy is of much importance for the learning-based control law since the closed-loop system may become unstable during the learning process under inappropriate initial settings and parameter selections. Some researchers add an additional stabilizing term to the NN update law [31], which increases the algorithm complexity to some extent. In this article, a PD-like controller easy for reconstruction of basis functions is chosen as the initial policy of the proposed online learning scheme.

IV. NUMERICAL SIMULATIONS

In this section, numerical simulations are carried out to show the effectiveness of the proposed control scheme. The target is freely tumbling on a Molniya orbit with orbital parameters listed in Table I. The target tumbling motion is determined by (8) and (9) with the inertia matrix $\mathbf{J}_t =$

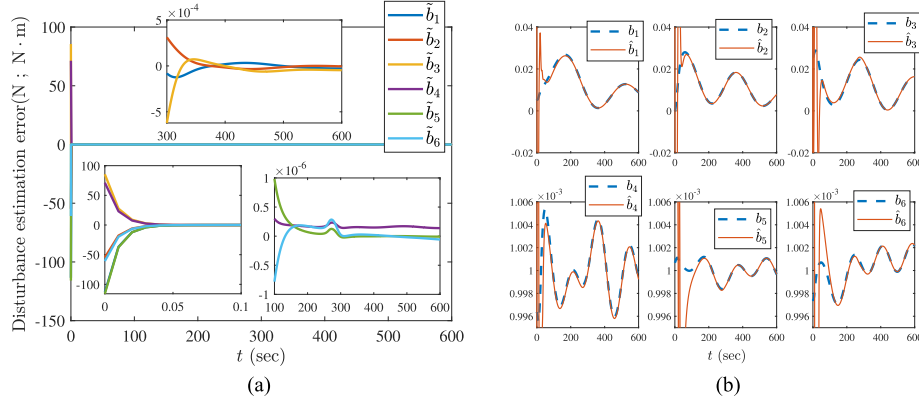


Fig. 6. Simulation results of the disturbance estimation. (a) DOB outputs. (b) Disturbances and their estimations.

TABLE I
Initial Orbital Elements of the Target

Orbital elements	Values	Units
Semimajor axis	26628	km
Eccentricity	0.7417	N/A
Inclination	63.4	deg
RAAN	0	deg
Argument of perigee	-90	deg
True anomaly	0	deg

$\text{diag}[55, 65, 58] \text{ kg} \cdot \text{m}^2$ and the initial conditions $\mathbf{q}_t(0) = [0, 0, 0, 1]^\top$ and $\boldsymbol{\omega}_{TN}^T(0) = [0.01, -0.01, 0.01]^\top \text{ rad/s}$. The mass and inertia parameters of the servicing spacecraft are $m_s = 100 \text{ kg}$ and $\mathbf{J}_s = \text{diag}[22, 20, 23] \text{ kg} \cdot \text{m}^2$, and its initial attitude and angular velocity are $\mathbf{q}_s(0) = [0.6968, -0.1593, -0.2237, -0.6626]^\top$ and $\boldsymbol{\omega}_{SN}^S(0) = [-0.01, 0.01, 0.02]^\top \text{ rad/s}$. The initial relative position and velocity of the servicing spacecraft with respect to the target are set to $\boldsymbol{\rho}(0) = [-144.92, 53.98, -10.42]^\top \text{ m}$ and $\mathbf{v}(0) = [0.257, 0.190, 0.195]^\top \text{ m/s}$. The thruster configuration parameters are set as $d_x = d_y = d_z = 1 \text{ m}$, and the thrust limit is $\gamma = 10 \text{ N}$. The servicing spacecraft is required to arrive at the anchoring point $\mathbf{p}_d = [-15, 0, 0]^\top \text{ m}$, while obeying both path constraint ($\alpha_m = 15^\circ$) and FOV constraint ($\beta_m = 30^\circ$).

A. Performance Validation

Sampling time step is $\Delta t = 0.01 \text{ s}$, DOB parameters are set as $k = 10$, $\chi = 0.04$, $\varepsilon = 0.3$, $m = 3$, $n = 5$, and $\mathbf{z}(0) = [20, 7, 40, 0.3, 0.5, 0.26]^\top$. The control parameters are taken as $\mathbf{Q} = \text{blkdiag}[\mathbf{Q}_\rho, \mathbf{Q}_v, \mathbf{Q}_q, \mathbf{Q}_\omega]$ with $\mathbf{Q}_\rho = \text{diag}[2, 1, 0.1]$, $\mathbf{Q}_v = 0.1\mathbf{I}_3$, $\mathbf{Q}_q = \mathbf{I}_3$, and $\mathbf{Q}_\omega = 0.1\mathbf{I}_3$, $\eta_1 = 0.2$, $\eta_2 = 0.05$, $k_1 = 0.1$, $k_2 = 1$, $\mathbf{P}_1 = 0.1\mathbf{I}$, and $\mathbf{P}_2 = 0.3\mathbf{I}$. The basis function is chosen as $\boldsymbol{\sigma} = [r_{e1}v_{e1}, r_{e2}v_{e2}, r_{e3}v_{e3}, q_{e1}\omega_{e1}, q_{e2}\omega_{e2}, q_{e3}\omega_{e3}, v_{e1}^2, v_{e2}^2, v_{e3}^2, \omega_{e1}^2, \omega_{e2}^2, \omega_{e3}^2]^\top$. The closed-loop responses are depicted in Fig. 5, where the relative position and velocity errors, attitude, and angular rate errors are asymptotically convergent. This shows that the servicing spacecraft successfully accomplishes the proximity operations with the freely tumbling target, while keeping its line of sight straight toward the target. The disturbance observation

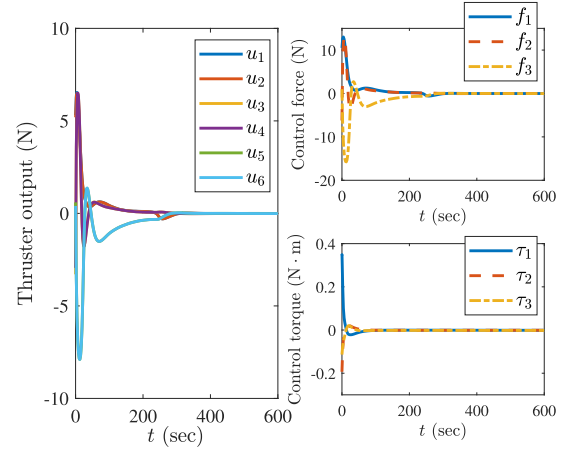


Fig. 7. Control inputs and thruster outputs.

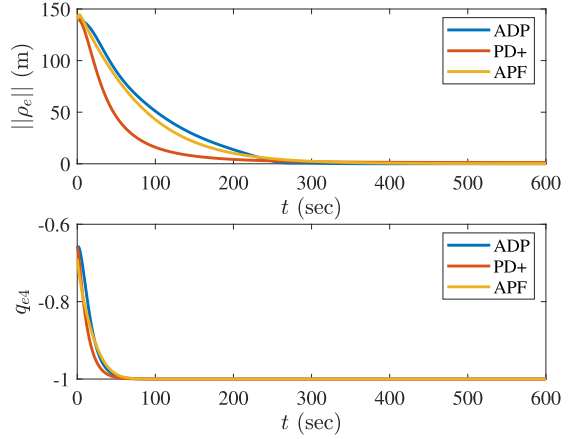


Fig. 8. Position and attitude tracking errors.

results are plotted in Fig. 6, where Fig. 6(a) shows the disturbance estimation error, and Fig. 6(b) compares the actual disturbances and their estimates. It is noted that larger gains are chosen for the DOB to obtain quick and precise disturbance estimation, which results in large observation errors at the initial stage, as shown in Fig. 6(a). As can be seen, the disturbance estimation errors rapidly converge to small residual sets around zero at about 0.1 s.

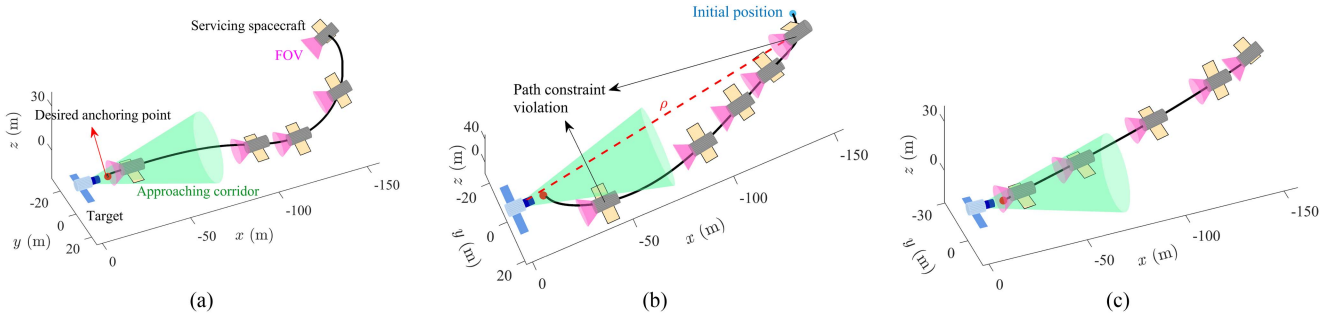


Fig. 9. Comparison results of 3-D motion trajectories. (a) Proposed controller. (b) PD+ controller. (c) APF-based controller.

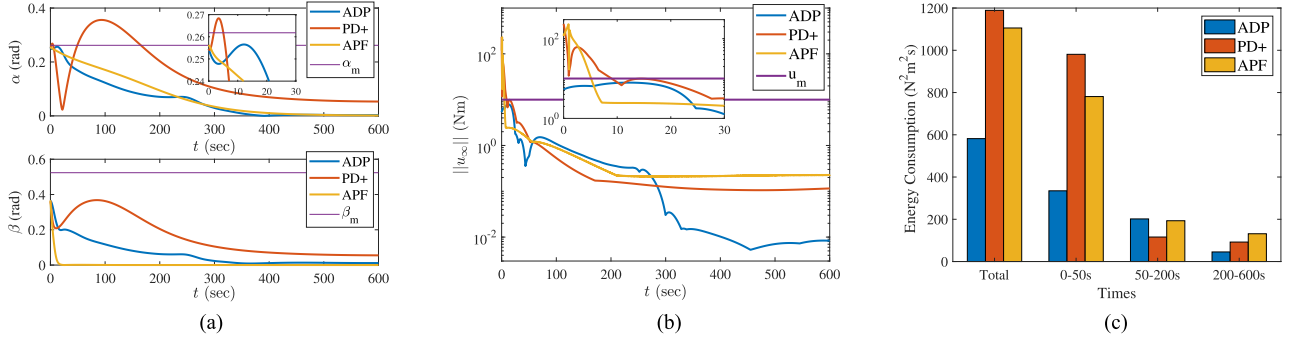


Fig. 10. Comparison results of motion constraints, control inputs, and energy consumptions. (a) Motion constraints. (b) Control input norm. (c) Energy consumption.

Fig. 7 depicts the control inputs, which are limited below the maximum allowable value $\gamma = 10$ N. The actuating forces and torques generated by the thruster pairs are shown in Fig. 7. Furthermore, the 3-D motion trajectory of the servicing spacecraft relative to the target in the frame $\mathcal{T}$ is shown in Fig. 9(a), where the path and FOV constraints are marked with green and purple cones. As can be clearly seen, both constraints are strictly satisfied during the proximity operations.

B. Comparison Results

For comparison purposes, two other methods are also simulated, one of which is the PD+ controller given by

$$\begin{aligned} \mathbf{f} &= -\mathbf{h}_p - \mathbf{\Gamma}_{p1}\boldsymbol{\rho}_e - \mathbf{\Gamma}_{d1}\mathbf{v}_e \\ \boldsymbol{\tau} &= -\mathbf{h}_a - \mathbf{\Gamma}_{p2}\mathbf{q}_e - \mathbf{\Gamma}_{d2}\boldsymbol{\omega}_e \end{aligned}$$

where $\mathbf{\Gamma}_{pi} = \text{diag}[\Gamma_{i1}, \Gamma_{i2}, \Gamma_{i3}]$ and $\mathbf{\Gamma}_{di} = \text{diag}[\Gamma_{i1}, \Gamma_{i2}, \Gamma_{i3}]$, $i = 1, 2, 3$. Another is the APF-based controller proposed in [33], which is formulated as

$$\begin{aligned} \mathbf{f} &= -\mathbf{R}_{ST}\boldsymbol{\Phi}_t(\hat{\mathbf{m}}_s + \boldsymbol{\beta}_t) \\ \boldsymbol{\tau} &= -\boldsymbol{\Phi}_r(\hat{\boldsymbol{\theta}} + \boldsymbol{\beta}_r) \end{aligned} \quad (58)$$

where $\boldsymbol{\Phi}_t \in \mathbb{R}^3$ and $\boldsymbol{\Phi}_r \in \mathbb{R}^{3 \times 6}$ are two known regressor matrices satisfying $\boldsymbol{\Phi}_t \mathbf{m}_s = -\mathbf{C}\mathbf{v}_e - \mathbf{G} + \mathbf{m}_s(\mathbf{N}_v^{-1}\mathbf{s}_t + k_1\mathbf{N}_v\mathbf{v}_e)$ and $\boldsymbol{\Phi}_r \boldsymbol{\theta} = \mathbf{h}_a + \mathbf{J}_s^{-1}(\mathbf{N}_\omega^{-1}\mathbf{s}_r + k_2\mathbf{N}_\omega\boldsymbol{\omega}_e)$, respectively, while $\hat{\mathbf{m}}_s + \boldsymbol{\beta}_t$ and $\hat{\boldsymbol{\theta}} + \boldsymbol{\beta}_r$ denote the estimates of unknown mass and inertia parameters. Refer to [33] for more details. For a fair comparison, all parameters are

tuned to ensure an almost identical settling time of position and attitude tracking errors, as shown in Fig. 8. The proposed controller, PD+ controller, and APF-based controller are abbreviated as ADP, PD+, and APF, respectively. The comparison result of 3-D motion trajectories in $\mathcal{T}$ is shown in Fig. 9, from which it can be seen that all the three controllers steer the servicing spacecraft to reach the desired docking/berthing port with its line of sight pointing toward the target. However, the PD+ controller violates path constraint, as shown in Fig. 9(b). To intuitively illustrate the constraint-handling ability of controllers, the time responses of α and β are plotted in Fig. 10(a). As can be seen, the proposed ADP controller and the APF-based controller ensure that $\alpha < \alpha_m = 15^\circ$ and $\beta < \beta_m = 30^\circ$, indicating that both path and FOV constraints are satisfied. However, under PD+ control scheme, α exceeds $\alpha_m = 15^\circ$ at about 2–5 s and 50–170 s, which is consistent with the 3-D trajectory in Fig. 9(b). In addition, the comparison results of control inputs are shown in Fig. 10(b). It is clear that the PD+ and APF-based controllers violate the input constraints, while the proposed ADP controller generates saturated control signals. Fig. 10(c) compares the energy consumption of the three controllers. It can be seen that the proposed ADP-based controller consumes the least energy, benefiting from performance optimization.

C. Simulation Scenario Under Pulse Modulation

Thruster pulse modulation is further considered to test the practical effectiveness of the proposed control method

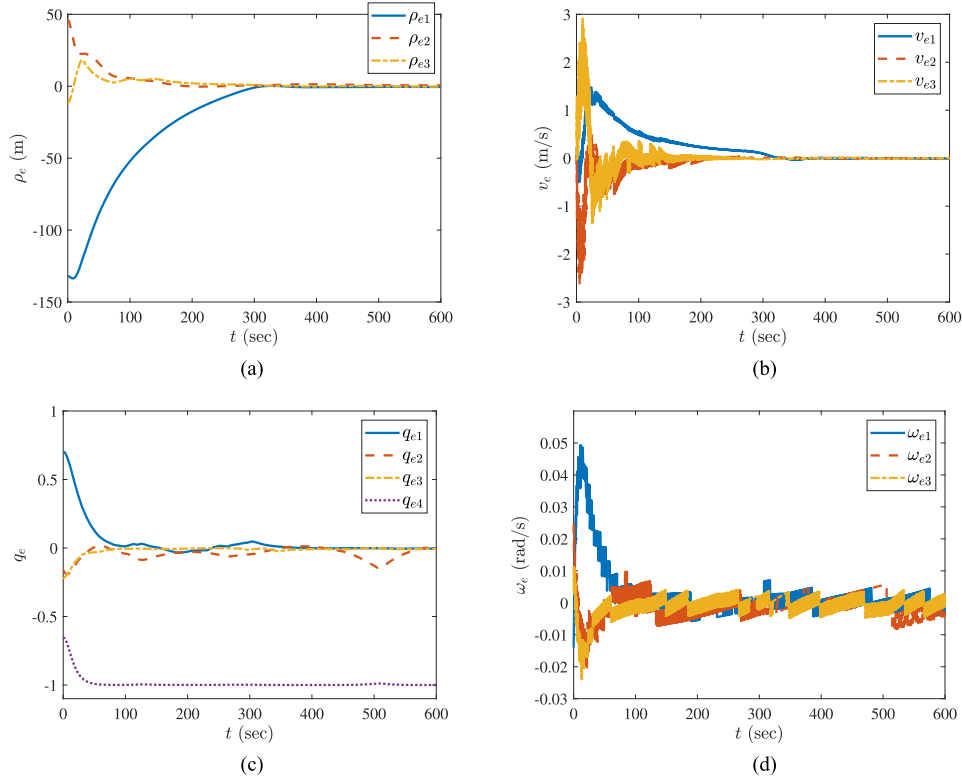


Fig. 11. Time responses with PWPF modulation. (a) Relative position errors. (b) Relative velocity errors. (c) Attitude errors. (d) Angular rate errors.

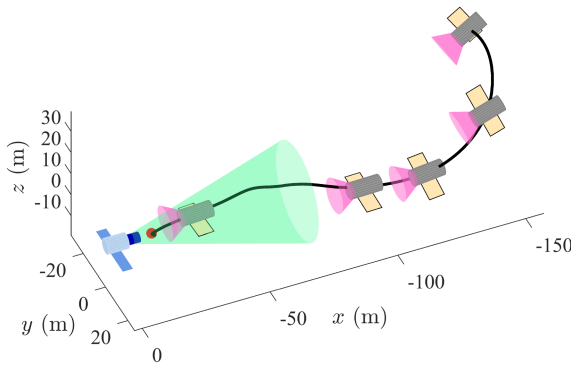


Fig. 12. 3-D motion trajectory under PWPF modulation.

in a more realistic environment. The pulswidth pulse-frequency (PWPF) modulator is adopted to transform the smooth input signals to ON-OFF signals. Interested readers may refer to [47] and [48] for more details on the PWPF modulator. The filter gain, time constant, switch-ON value, and switch-OFF value of the PWPF modulator are set as 6, 0.7, 0.1, and 0.5, respectively. Simulation results are shown in Figs. 11–13. It can be seen from Fig. 11 that the steady-state accuracy decreases in the practical scenario. But nonetheless, such performance degradation is still acceptable for certain missions. The chattering phenomenon observed in the velocity and angular velocity tracking errors is actually caused by the ON-OFF nature of PWPF. However, motion constraints are still satisfied during the proximity operations, as shown in Fig. 12. Fig. 13 depicts the impulsed control signals of six thruster pairs, wherein u_i

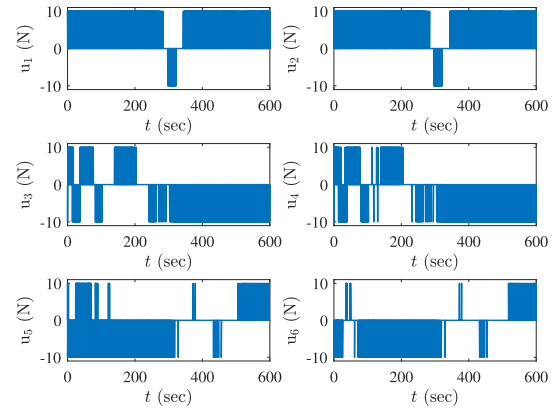


Fig. 13. Time histories of impulse control forces.

corresponds to the output of thruster pair $\{T_i, L_i\}$ (detailed thruster configuration is plotted in Fig. 3). It is shown that thrusters are switched ON and OFF repeatedly to drive the servicing spacecraft, performing relative position and attitude maneuvers. The positive (+10 N) and negative (−10 N) thrusts correspond to two opposite firing directions, which means either T_i or L_i is ON. When the two thrusters $\{T_i, L_i\}$ are both OFF, the output u_i of this thruster pair is zero.

V. CONCLUSION

This article presented a learning-based robust optimal control scheme for spacecraft proximity operations subject to motion and input constraints, as well as external disturbances, using the ADP technique. Motion and input

constraints as well as the unwinding problem were simultaneously solved by introducing a nonquadratic term and two barrier functions into the performance index. A finite-time disturbance observer is designed to learn environmental disturbances accurately, which enhances the robustness of the proposed learning algorithm. Based on the disturbance observation, two NN approximators were tuned online to yield a near-optimal saturated control policy. The derived controller was shown to achieve the boundedness of system states and network weight estimation errors, while ensuring the satisfaction of both motion and input constraints. Finally, simulation results demonstrated the effectiveness and performance of the proposed method. Our future work will focus on achieving model-free pose control for spacecraft engaged in constrained proximity operations.

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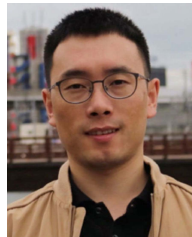
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